Model Analysis Report

Generated: March 04, 2026

Dataset Information

Number of timesteps: 136214

Spatial grid: 50 latitudes × 24 MLTs

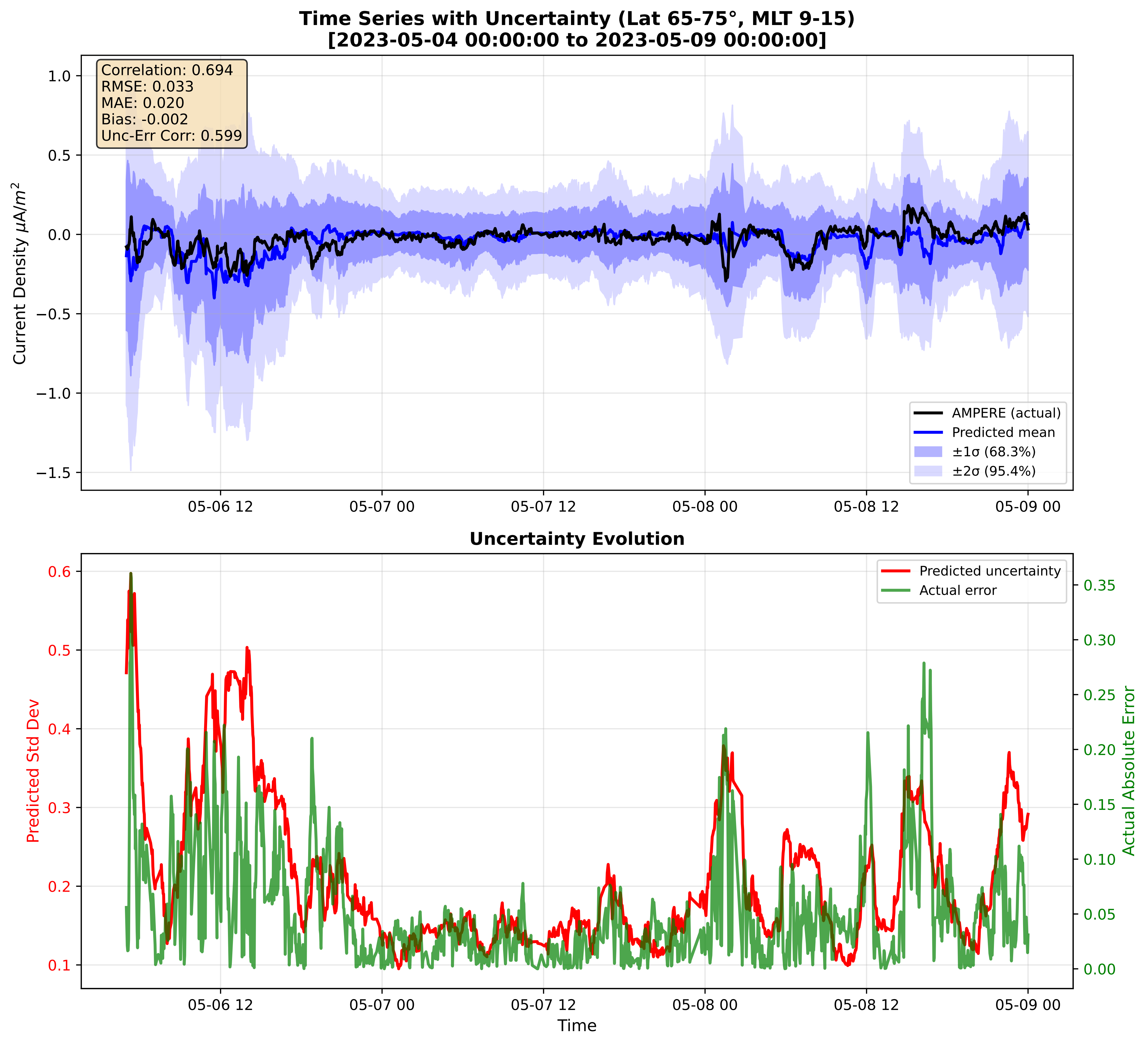
Time range: 2019-11-01 00:00:00 to 2024-01-31 23:56:00

Uncertainty available: Yes

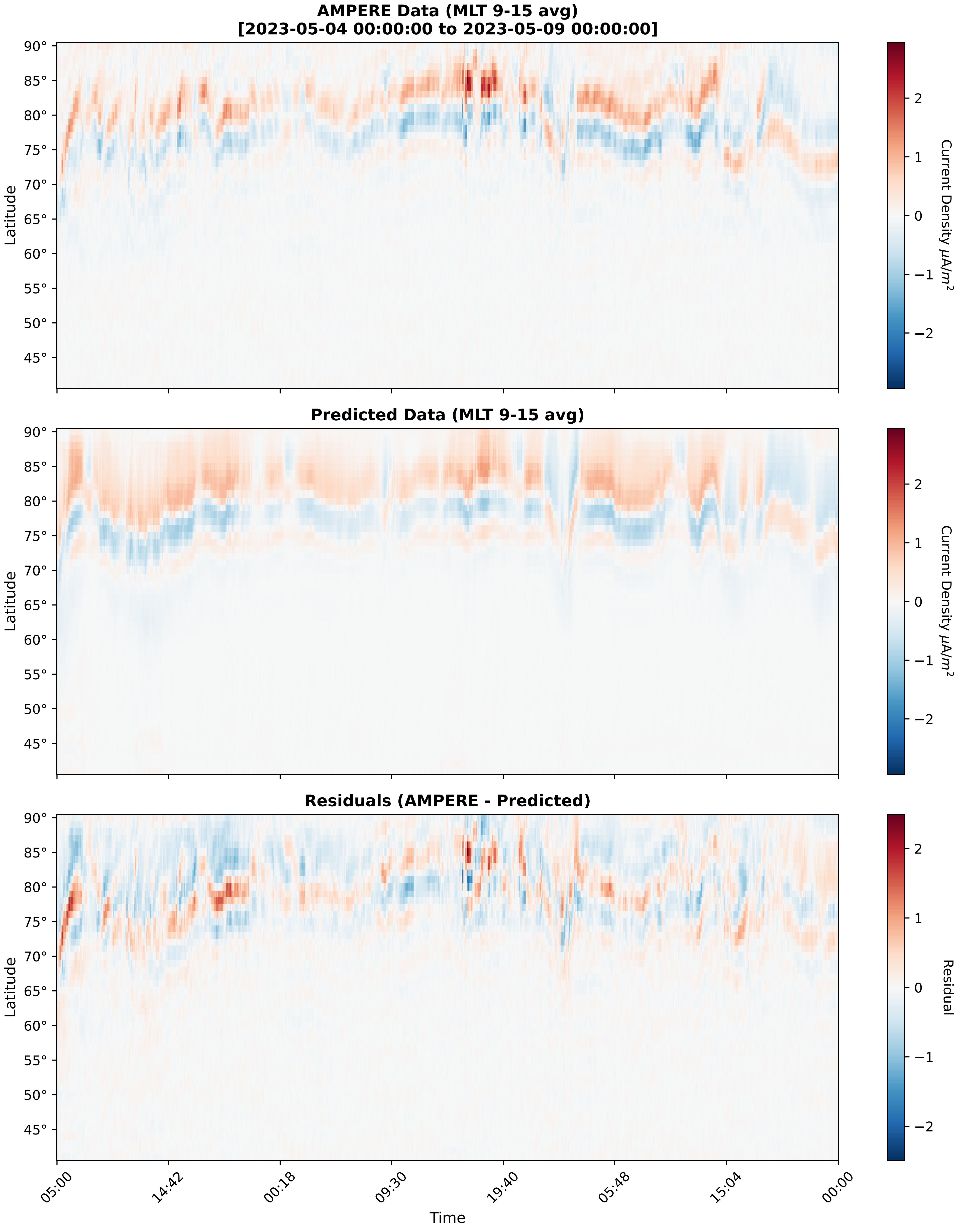
Analysis location: Lat 65-75°, MLT 9-15

Plot time range: 2023-05-04 00:00:00 to 2023-05-09 00:00:00

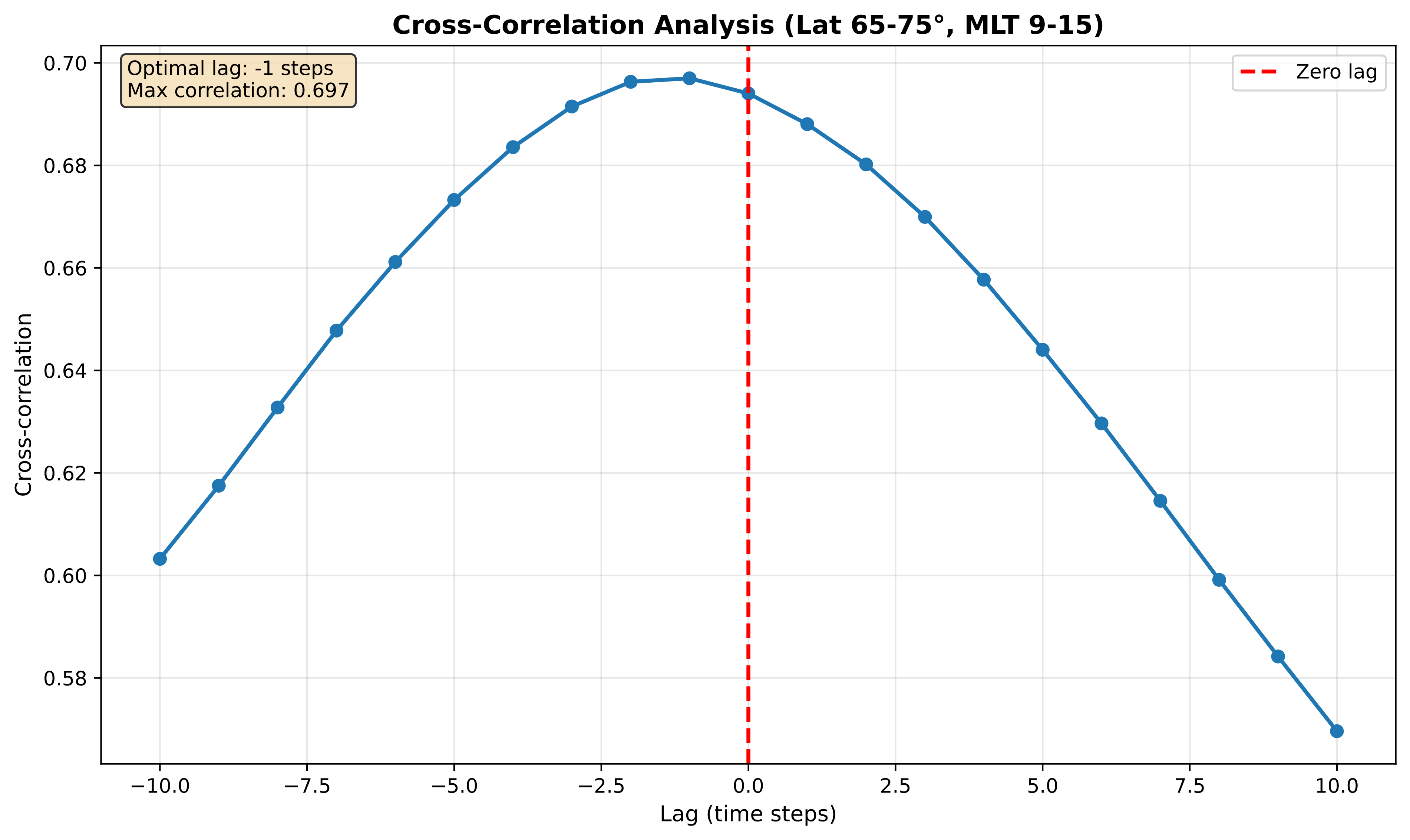
This report contains comprehensive validation metrics for geomagnetic model predictions

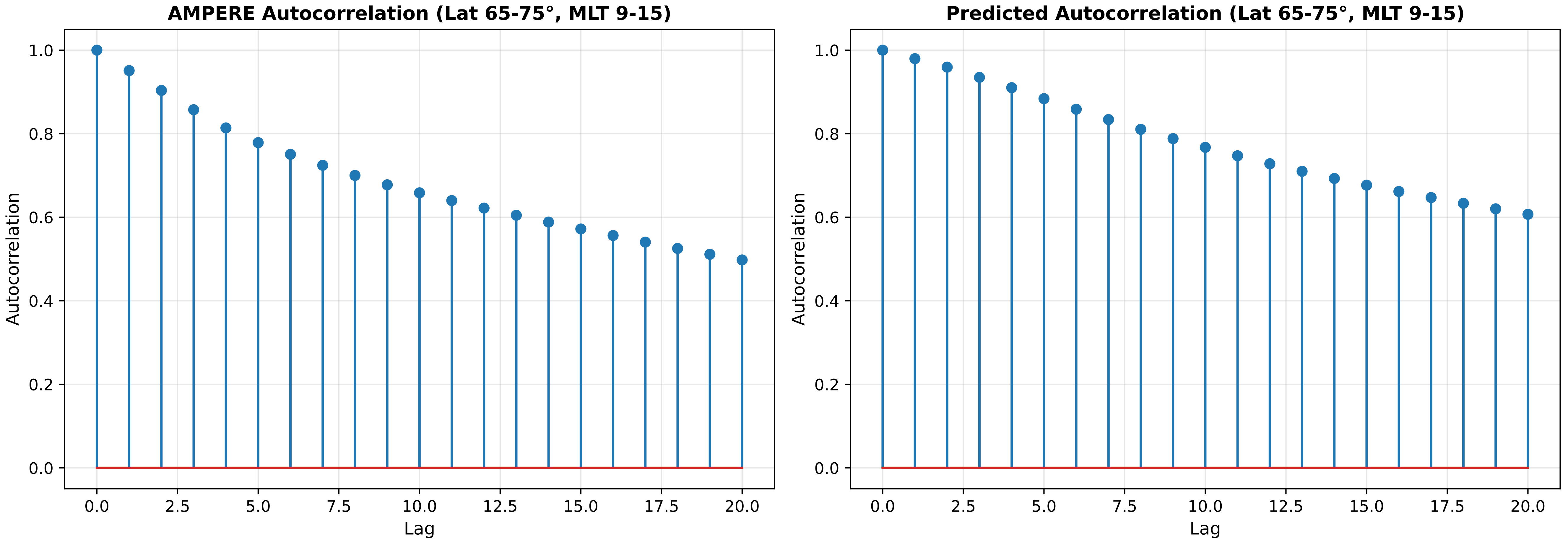


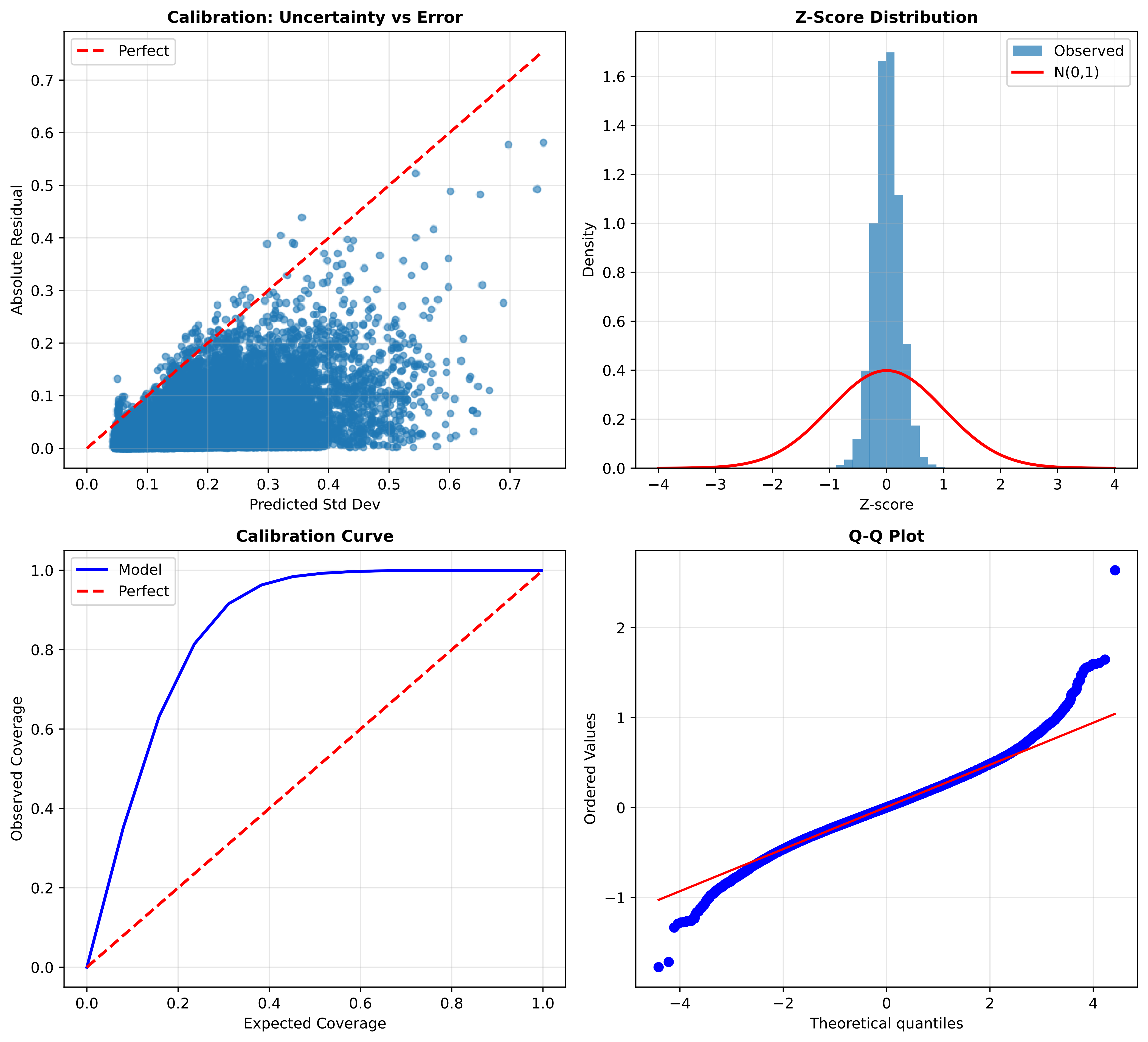
Time Series with Uncertainty (Lat 65-75°, MLT 9-15)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]
Correlation: 0.694
RMSE: 0.033
MAE: 0.020
Bias: -0.002
Unc-Err Corr: 0.599
Current Density $\mu A/m^2$
AMPERE (actual)
Predicted mean
±1σ (68.3%)
±2σ (95.4%)
05-06 12
05-07 00
05-07 12
05-08 00
05-08 12
05-09 00
Uncertainty Evolution
Predicted uncertainty
Actual error
Predicted Std Dev
Actual Absolute Error
Time

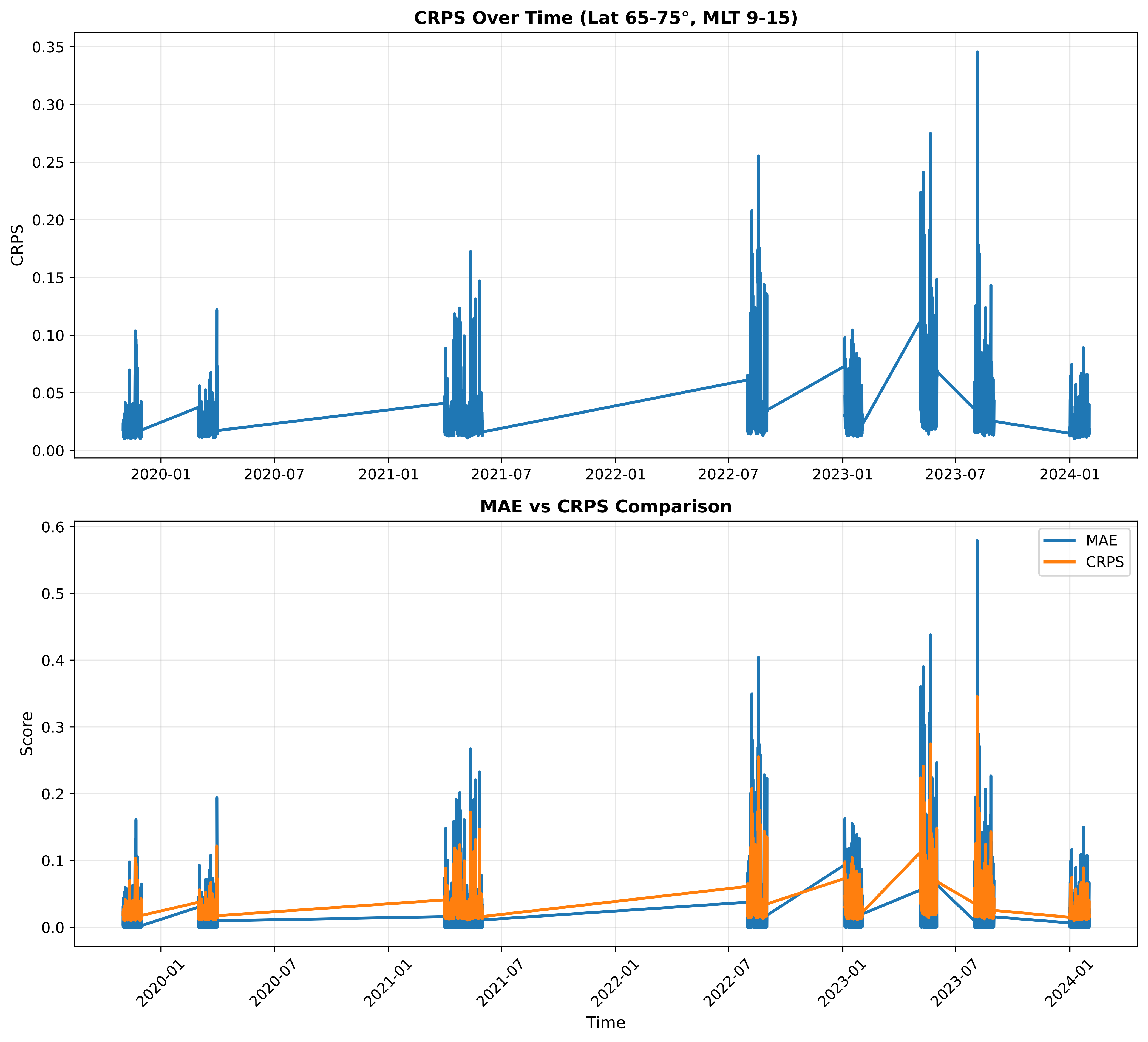


AMPERE Data (MLT 9-15 avg)
[2023-05-04 00:00:00 to 2023-05-09 00:00:00]
Predicted Data (MLT 9-15 avg)
Residuals (AMPERE - Predicted)
Latitude
90°
85°
80°
75°
70°
65°
60°
55°
50°
45°
Current Density $\mu A/m^2$
Residual
Time
05:00
14:42
00:18
09:30
19:40
05:48
15:04
00:00

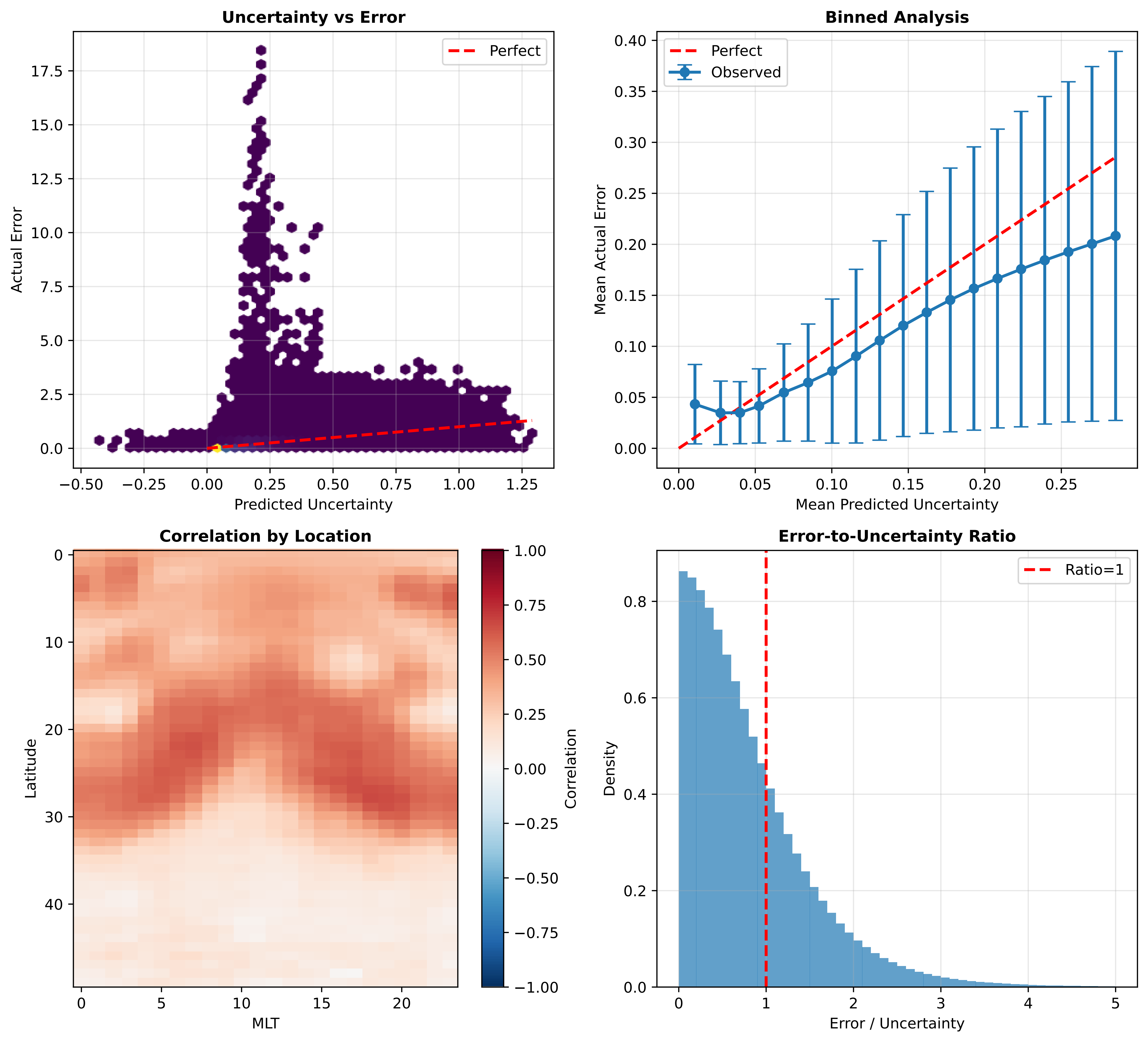

Cross-Correlation Analysis (Lat 65-75°, MLT 9-15)
Optimal lag: -1 steps
Max correlation: 0.697
Zero lag
Cross-correlation
Lag (time steps)
0.70
0.68
0.66
0.64
0.62
0.60
0.58
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

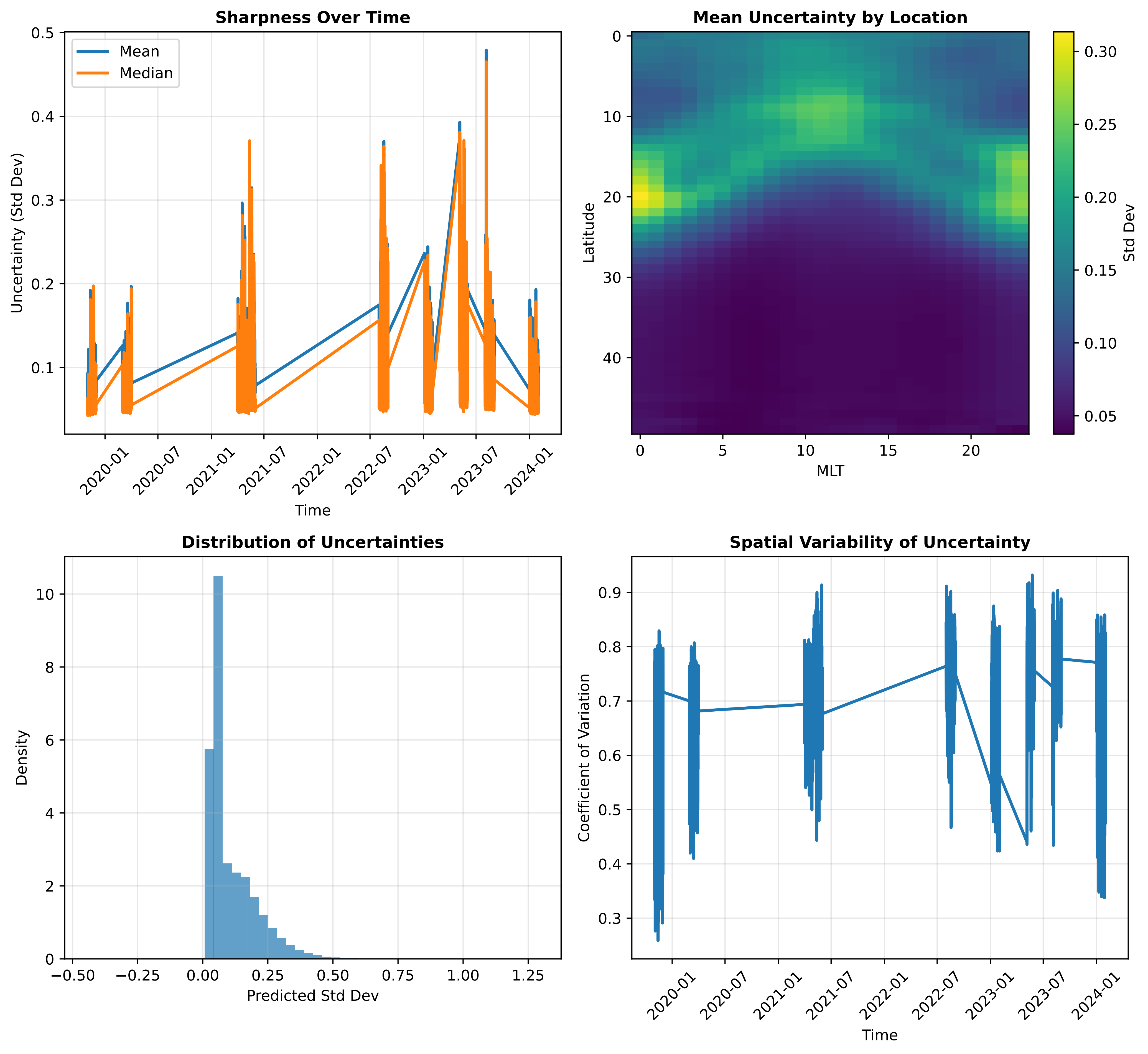

AMPERE Autocorrelation (Lat 65-75°, MLT 9-15)
Predicted Autocorrelation (Lat 65-75°, MLT 9-15)
Autocorrelation
Lag
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Calibration: Uncertainty vs Error
Perfect
Absolute Residual
Predicted Std Dev
Z-Score Distribution
Observed
N(0,1)
Density
Z-score
Calibration Curve
Model
Perfect
Observed Coverage
Expected Coverage
Q-Q Plot
Ordered Values
Theoretical quantiles



CRPS Over Time (Lat 65-75°, MLT 9-15)
CRPS
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
2020-01
2020-07
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
MAE vs CRPS Comparison
MAE
CRPS
Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Time


Uncertainty vs Error
Perfect
Actual Error
Predicted Uncertainty
Binned Analysis
Perfect
Observed
Mean Actual Error
Mean Predicted Uncertainty
Correlation by Location
Latitude
MLT
Correlation
Error-to-Uncertainty Ratio
Ratio=1
Density
Error / Uncertainty


Sharpness Over Time
Mean
Median
Uncertainty (Std Dev)
Time
Mean Uncertainty by Location
Latitude
MLT
Std Dev
Distribution of Uncertainties
Density
Predicted Std Dev
Spatial Variability of Uncertainty
Coefficient of Variation
Time

Numerical Metrics Summary

Deterministic Performance:

Point Correlation: 0.694

Point RMSE: 0.033

Point MAE: 0.020

Point Bias: -0.002

Uncertainty-Error Corr: 0.599

Optimal Lag: -1 timesteps

Max Cross-Correlation: 0.697

Uncertainty Quantification:

Coverage within 1σ: 99.9% (target: 68.3%)

Coverage within 2σ: 100.0% (target: 95.4%)

Z-score mean: 0.008 (target: 0)

Z-score std: 0.235 (target: 1)

Mean CRPS: 0.0275

CRPS/MAE Ratio: 1.347

Uncertainty-Error Correlation: 0.567

Median Error/Uncertainty Ratio: 0.639

Executive Summary

Point Performance:

Point prediction at Lat 65-75°, MLT 9-15 shows moderate correlation (r=0.694) with RMSE=0.033. Model exhibits negative bias of 0.002. Uncertainty-error correlation is 0.599, indicating strong awareness of prediction quality.

Spatial Performance:

Spatial analysis at MLT 9-15 shows mean RMSE of 0.091 across all latitudes. Maximum errors (0.236) occur at high latitudes.

Temporal Alignment:

Model leads observations by 1 timesteps (anticipates changes). Maximum achievable correlation is 0.697.

Temporal Dynamics:

Autocorrelation analysis shows decorrelation timescales of 20 timesteps for observations and 20 timesteps for predictions. Model captures temporal persistence well.

Uncertainty Quality:

Uncertainty calibration is poor: 99.9% within 1σ (expected 68.3%), 100.0% within 2σ (expected 95.4%). Model is underconfident (uncertainties too wide). Z-score statistics: mean=0.008 (ideal: 0), std=0.235 (ideal: 1).
CRPS/MAE = 1.35; Uncertainty does not add value beyond point predictions. Mean CRPS=0.0275, Mean MAE=0.0204.
Uncertainty-error correlation is 0.567, indicating weak relationship. Median error/uncertainty ratio is 0.639 (underconfident (uncertainties too large)). Model shows moderate awareness of uncertainty patterns.

Uncertainty Characteristics:

Model sharpness: mean uncertainty=0.1073, median=0.0623. Temporal variability (CV=0.365) suggests condition-dependent uncertainty. Spatial variability (CV=0.611) indicates location-dependent confidence.